Campus Lost & Found

A DevOps capstone: from a git push to a watched Kubernetes app



Divijaa · github.com/Divii2205/campus-lost-and-found

Project overview

THE APP

Lost something?

Search the board, then post a Lost report with the place and a contact.

Found something?

Post a Found report so the owner knows where it is.

Back with owner

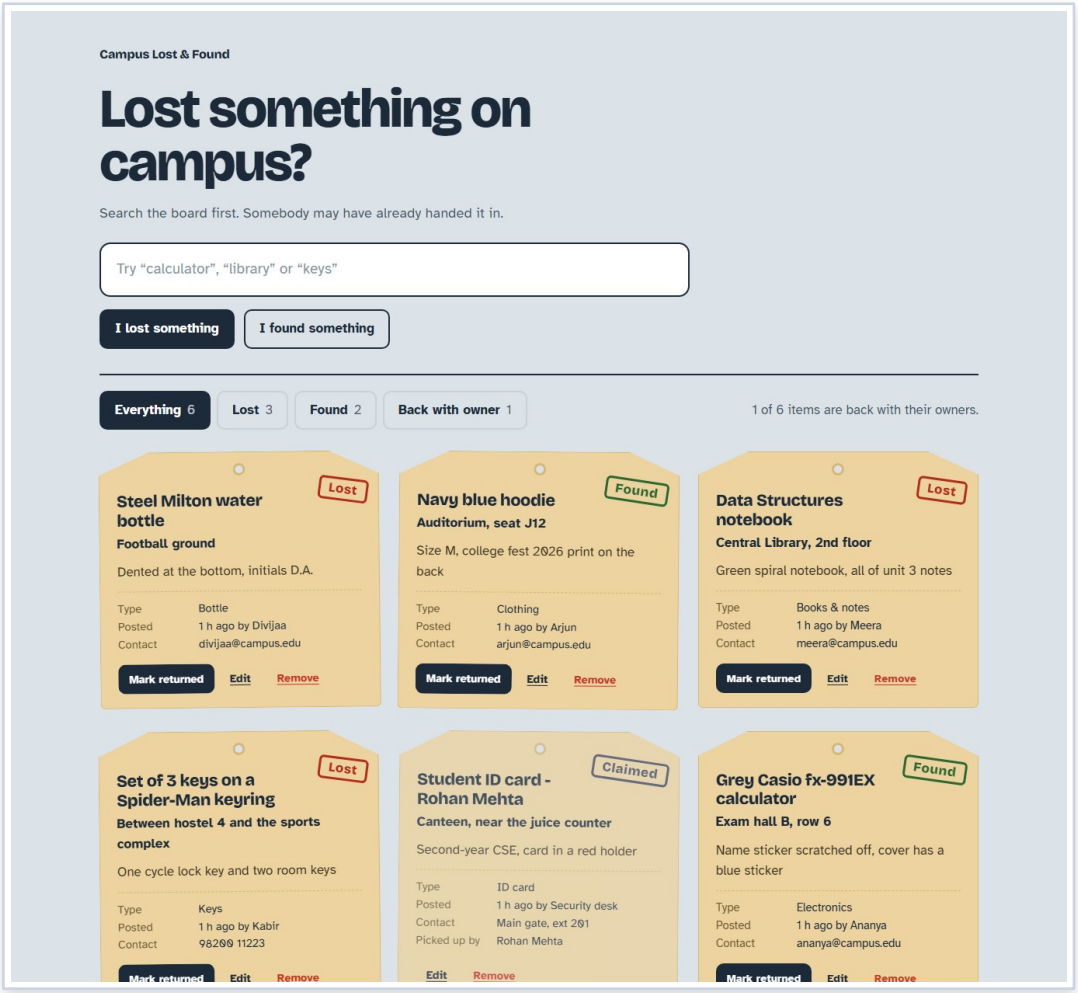
Mark it returned and it gets a Claimed stamp.

Under the hood

FastAPI, 7 item endpoints, PostgreSQL, 11 tests.

What I built around it

Every change is tested, scanned, packed into a SHA tagged image, deployed from Git by Argo CD and watched with Prometheus and Grafana.



The real app: each report is a paper tag with an ink stamp

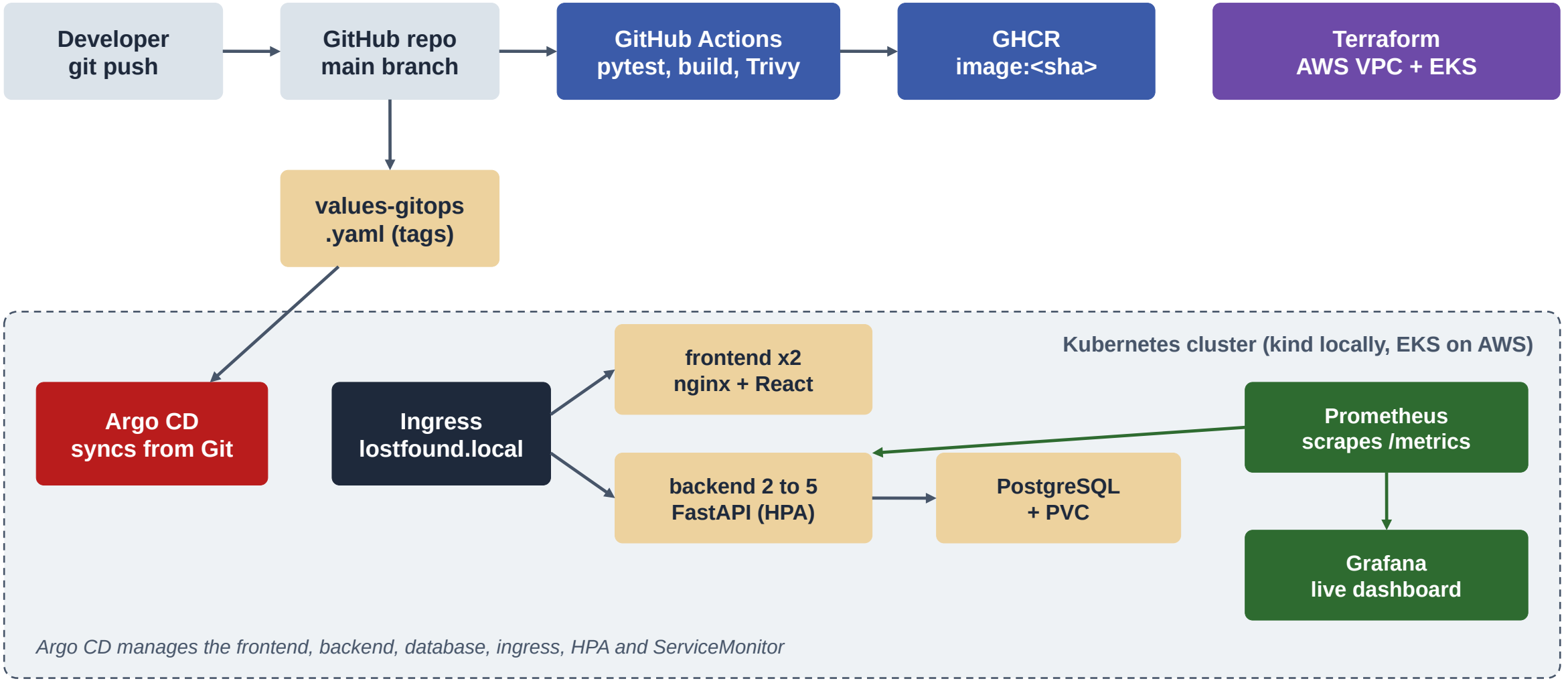
Tech stack

TOOLS

App	React 18 + Vite, FastAPI, SQLAlchemy, PostgreSQL 16, Alembic
Tests	pytest with a throwaway SQLite database per test
Containers	Docker non-root images, multi-stage frontend, Docker Compose
CI/CD	GitHub Actions, GitHub Container Registry, commit SHA tags
Security	Trivy image scan gate, gitleaks secret scan
Infrastructure	Terraform on AWS: VPC, subnets, IGW, EKS, node group, IAM
Kubernetes	kind, Helm 3, ingress-nginx, metrics-server, HPA
GitOps	Argo CD with auto sync, prune and self heal
Observability	kube-prometheus-stack: Prometheus, ServiceMonitor, Grafana

Architecture

BIG PICTURE



CI/CD flow

EVERY PUSH

1 test
pytest -v (11 tests)
npm ci and build

→

2 build
backend and frontend
images, tag = SHA

→

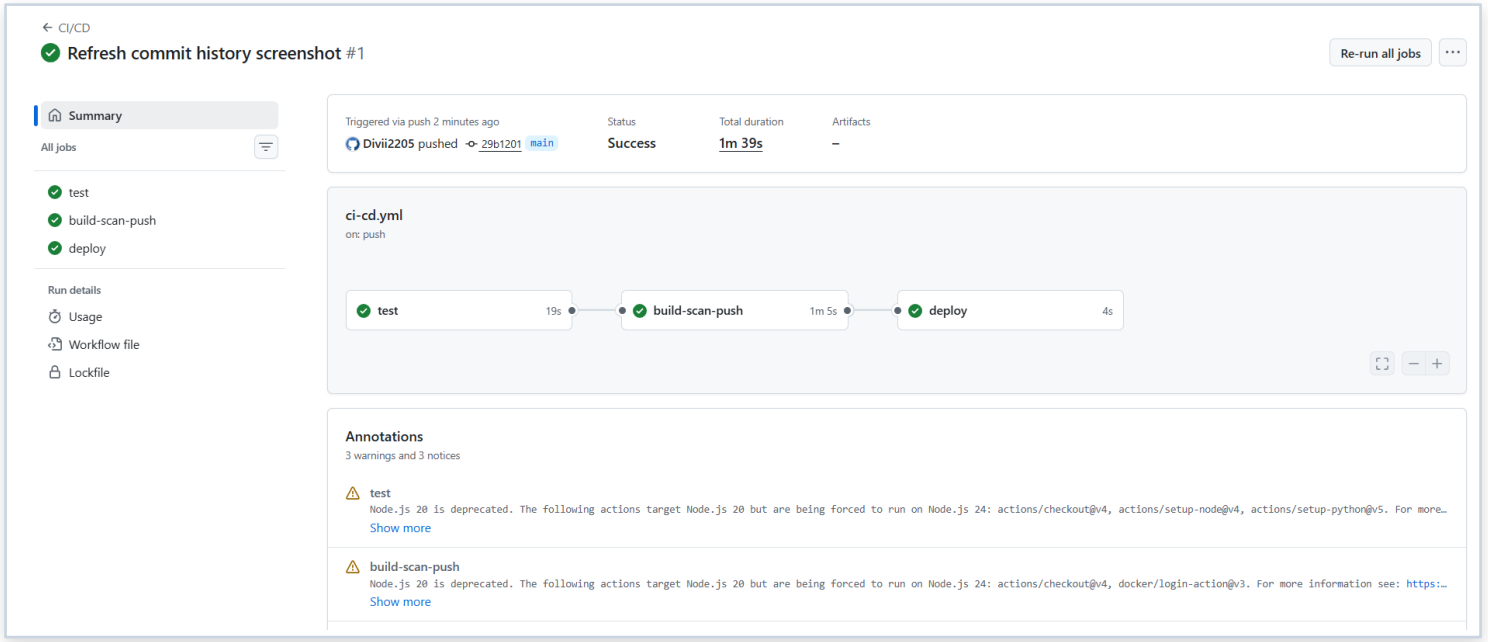
3 Trivy
fail on fixable
HIGH or CRITICAL

→

4 push
both images
to GHCR

→

5 deploy
Argo CD syncs the
new tag from Git



- Runs on every push and pull request to main
- A failed test or scan stops the pipeline, so no bad image reaches GHCR
- Image tags are the commit SHA, never latest

Security: a real Trivy finding

FIXED

Trivy image scan - backend BEFORE the fix

```
$ docker run aquasec/trivy:0.65.0 image --severity HIGH,CRITICAL --ignore-unfixed --exit-code 1 ghcr.io/divii2205/lostfound-backend:<sha>
Report Summary
```

Target	Type	Vulnerabilities	Secrets
ghcr.io/divii2205/lostfound-backend:c212244da9f86539501793e824ec01e783419fbe (debian 13.7)	debian	0	-
Python	python-pkg	4	-
usr/local/lib/python3.12/site-packages/SQLAlchemy-2.0.36.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/alembic-1.14.0.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/annotated_doc-0.0.5.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/annotated_types-0.8.0.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/anyio-4.15.1.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/certifi-2026.7.22.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/click-8.5.0.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/fastapi-0.142.4.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/greenlet-3.5.6.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.12/site-packages/h11-0.16.0.dist-info/METADATA	python-pkg	0	-
... (60 lines skipped) ...			
usr/local/lib/python3.12/site-packages/websockets-17.2.dist-info/METADATA	python-pkg	0	-

```
Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)


Python (python-pkg)
===================
Total: 4 (HIGH: 4, CRITICAL: 0)
```

Library	Vulnerability	Severity	Status	Installed Version	Fixed Version	Title
msgpack	GHSA-6v7p-g79w-8964	HIGH	fixed	1.1.2	1.2.1	MessagePack for Python: Out-of-bounds read / crash on Unpacker reuse after a... https://github.com/advisories/GHSA-6v7p-g79w-8964
setuptools	CVE-2025-47273			70.3.0	78.1.1	setuptools: Path Traversal Vulnerability in setuptools PackageIndex https://avd.aquasec.com/nvd/cve-2025-47273
urllib3	CVE-2026-97687			2.7.0	2.8.0	urllib3: urllib3: Traffic interception via HTTPS proxy TLS configuration override https://avd.aquasec.com/nvd/cve-2026-97687
	CVE-2026-97689					urllib3: urllib3: Denial of Service via unbounded memory allocation in chunk parser... https://avd.aquasec.com/nvd/cve-2026-97689

```
>>> exit code 1: pipeline gate FAILS
```

Before: 4 HIGH problems in the first backend image

CVE-2025-47273

A path traversal bug in setuptools 70.3.0. It came from pip and setuptools left in the image, not from the app.

The fix

Remove pip and setuptools after pip install. The app never needs them while it runs.

Trivy image scan - both images AFTER the fix

```
$ trivy image --severity HIGH,CRITICAL --ignore-unfixed --exit-code 1 ghcr.io/divii2205/lostfound-backend:040621990b8dd1cf6f6225bde4ba1f3c49153667
| ghcr.io/divii2205/lostfound-backend:040621990b8dd1cf6f6225bde4ba1f3c49153667     | debian |       0       |    -    |
  OS packages       : 0 targets with HIGH/CRITICAL
  language packages: 37 scanned, 0 with HIGH/CRITICAL
>>> trivy exit code 0 -> gate PASSES

$ trivy image --severity HIGH,CRITICAL --ignore-unfixed --exit-code 1 ghcr.io/divii2205/lostfound-frontend:040621990b8dd1cf6f6225bde4ba1f3c49153667
| ghcr.io/divii2205/lostfound-frontend:040621990b8dd1cf6f6225bde4ba1f3c49153667 | alpine |       0       |    -    |
  OS packages       : 0 targets with HIGH/CRITICAL
  language packages: 0 scanned, 0 with HIGH/CRITICAL
>>> trivy exit code 0 -> gate PASSES
```

After: 0 HIGH or CRITICAL in both images. gitleaks also found no secrets in Git.

Infrastructure with Terraform

AWS

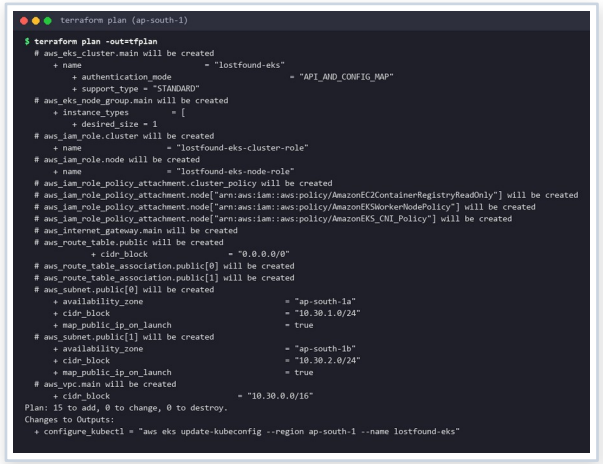
15
resources

2
public subnets

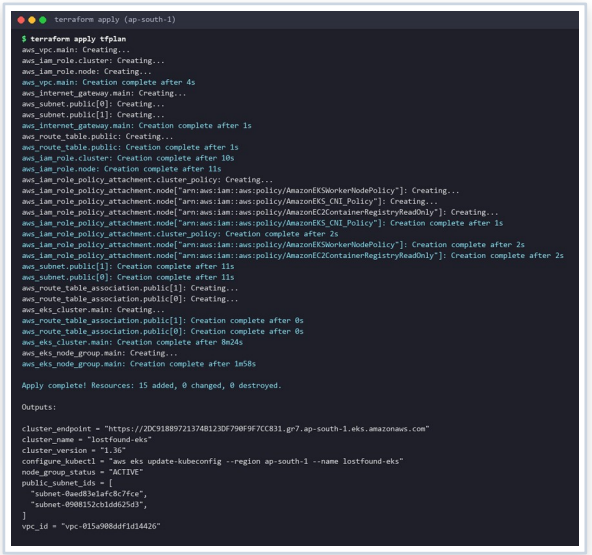
12 min
to apply

$0
left after destroy

- main.tf: VPC 10.30.0.0/16, 2 public subnets in 2 zones, internet gateway, route table
- eks.tf: IAM roles, EKS cluster lostfound-eks, managed node group (t3.small)
- variables.tf and tfvars.example: no keys saved anywhere
- No NAT gateway, to keep the cost low



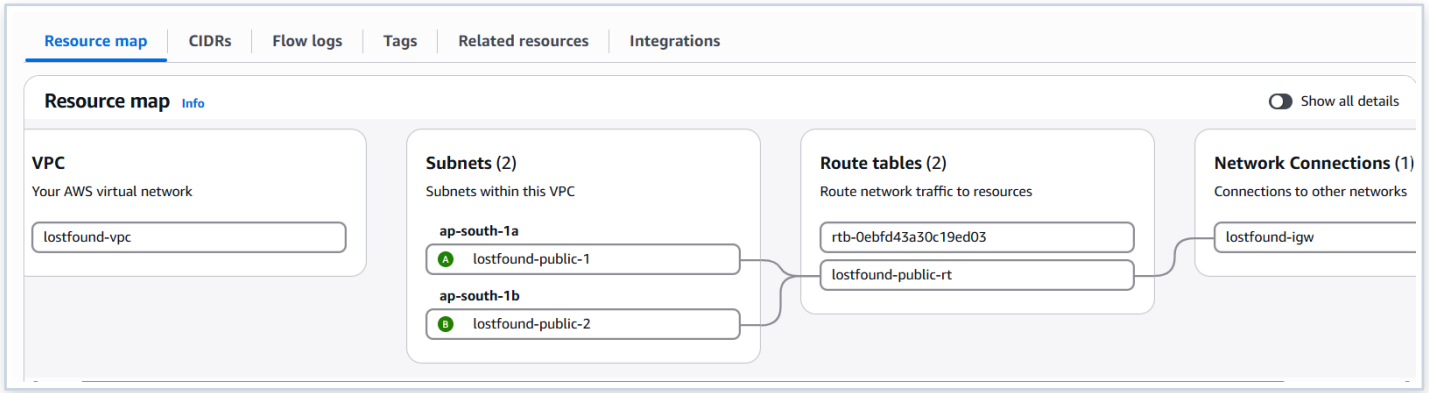
terraform plan: 15 to add



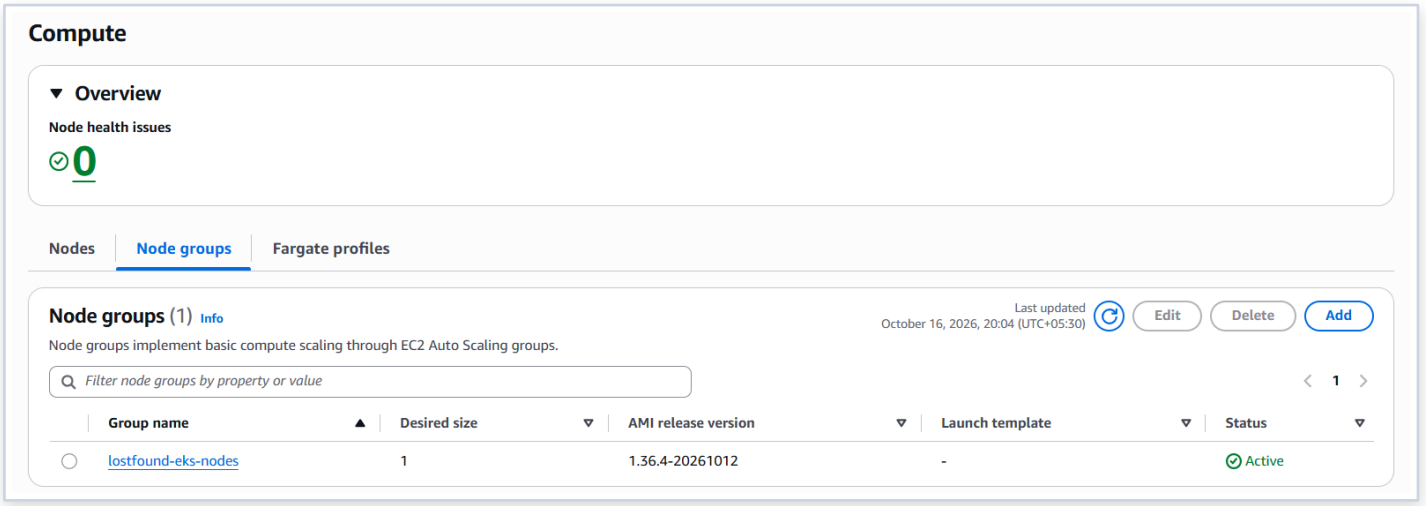
terraform apply: 15 added

Proof on AWS: VPC and EKS

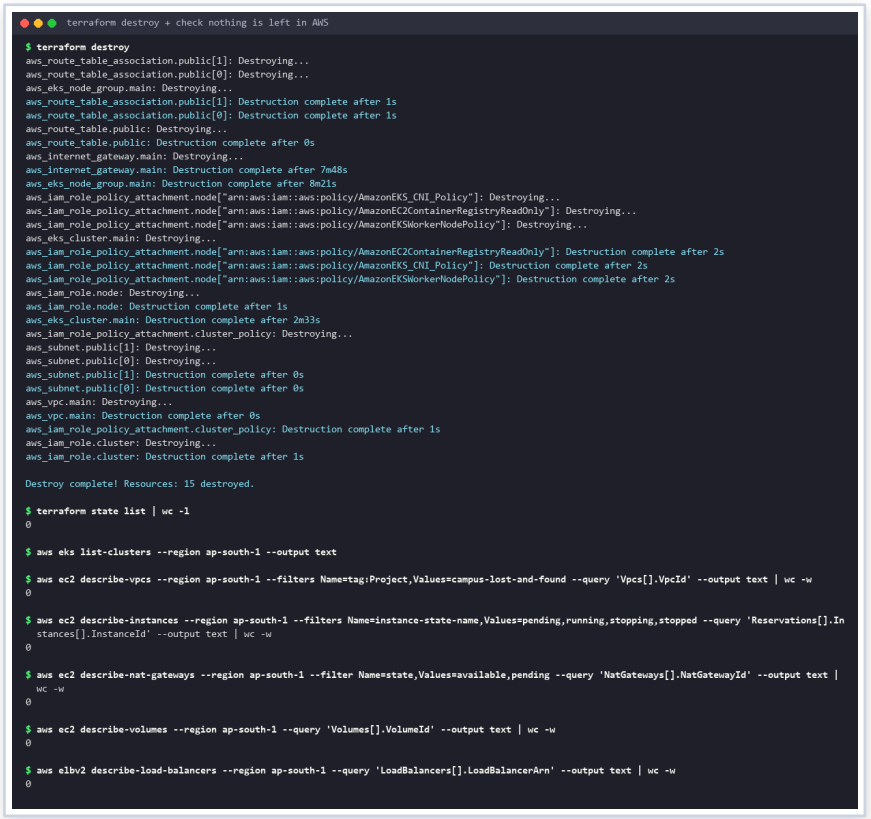
BUILT



AWS Console: the VPC with 2 public subnets, route table and internet gateway



AWS Console: EKS cluster lostfound-eks with node group Active



Then terraform destroy: nothing left running

Kubernetes setup with Helm

HELM

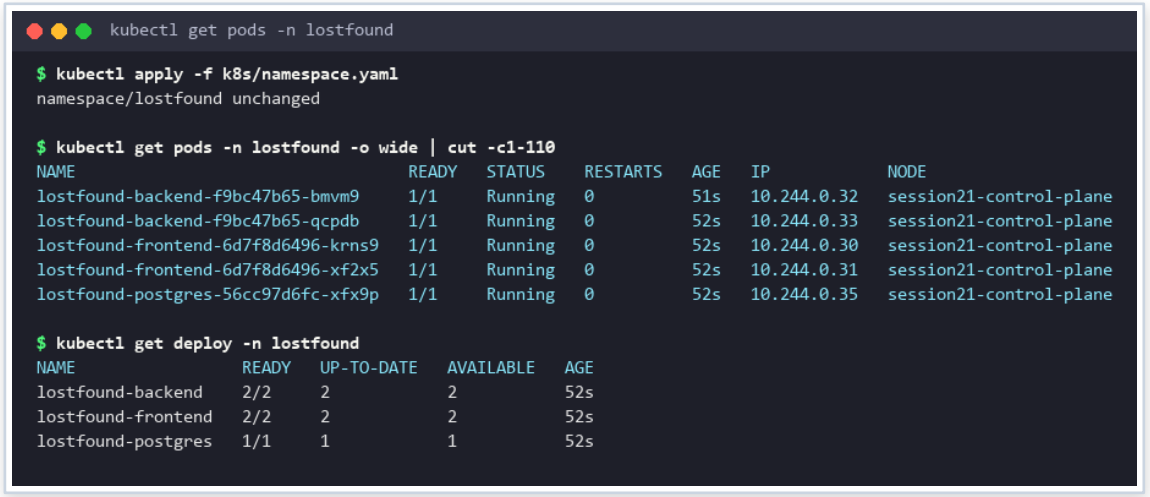
Deployments
frontend x2, backend x2, PostgreSQL with a 1Gi PVC

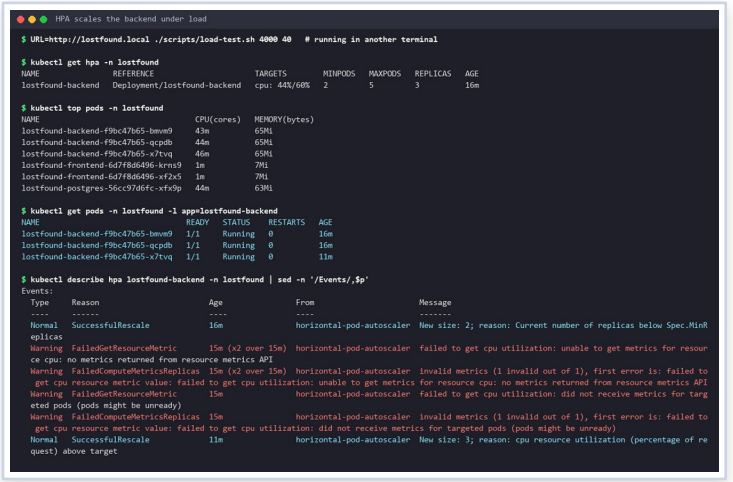
Services + Ingress
ClusterIP Services. / goes to frontend, /api to backend

Probes + initContainer
startup, readiness, liveness. wait-for-db stops crash loops

Config
ConfigMap for settings, Secret for the database password

HPA
backend grows from 2 to 5 pods at 60% CPU





GitOps with Argo CD

SYNCED

git push
code change

↓

CI green
tested, scanned, in GHCR

↓

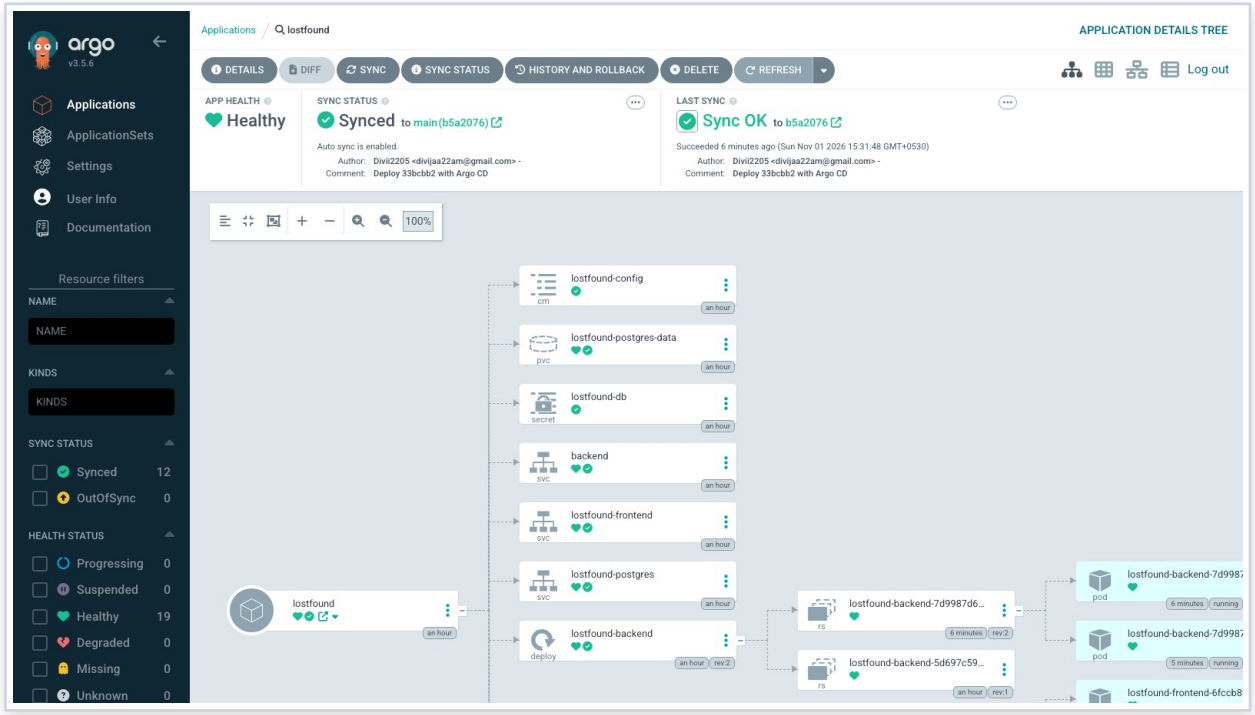
new SHA in Git
values-gitops.yaml

↓

Argo CD sync
Synced + Healthy

↓

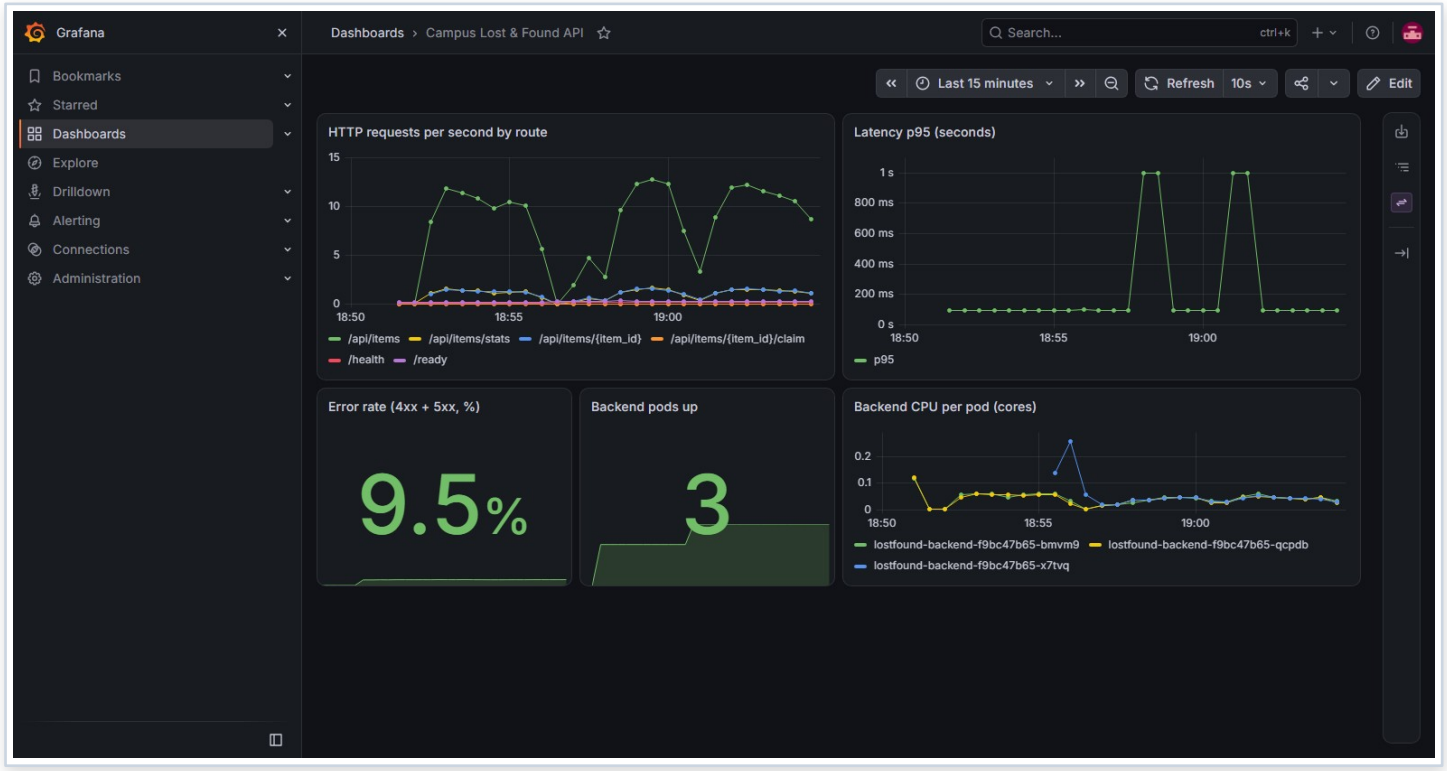
new pods
rolling update



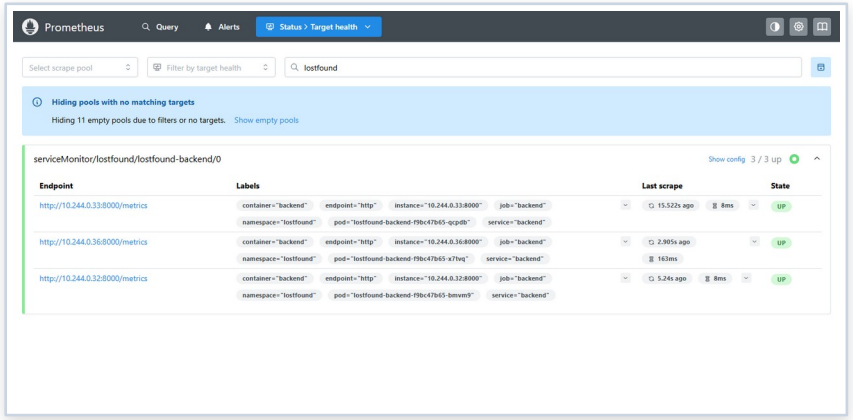
- argocd/application.yaml points Argo CD at helm/lostfound on main
- Auto sync, prune and self heal: Git is the only way to change the cluster
- A release is a commit, so every deploy has a history and an easy rollback

Observability

WATCHED



Grafana: requests per second, p95 latency, error rate, pods up, CPU per pod



Prometheus Targets: every backend pod UP

- Backend exposes /metrics in the Prometheus format
- A ServiceMonitor tells Prometheus to read it every 15 seconds
- The dashboard is a ConfigMap, so it is code too

Problems I hit and fixed

RUNBOOK

Backend crash loop

Why: Backend ran migrations before PostgreSQL was ready.

Fix: wait-for-db initContainer with pg_isready

Trivy gate failed

Why: pip and setuptools carried 4 HIGH CVEs.

Fix: remove them from the runtime image

ImagePullBackOff

Why: A tag that does not exist in the registry.

Fix: describe pod, then set a real SHA tag

Service with no endpoints

Why: Selector app=lostfound-api matched no pods.

Fix: fix the selector to the pod label

Key learnings

LEARNED

1. **Ready is not alive**
   /health says the process runs. /ready also checks the database, so traffic only goes to pods that can help.

2. **Start order matters**
   Compose has depends_on. Kubernetes has no start order, so an initContainer does the waiting.

3. **Most CVEs are leftovers**
   The app libraries were clean. The holes came from build tools left in the image.

4. **SHA tags make deploys traceable**
   Every pod maps to one commit, and a rollback is one revert in Git.

5. **Cloud costs money while it exists**
   No NAT gateway, one small node and destroy right after use kept AWS under a dollar.

CLAIMED

Thank you

Repository: github.com/Divii2205/campus-lost-and-found

Demo video: demo/demo-video.mp4

Every screenshot in these slides is real output from this project.

Divijaa